Junteng Liu

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Research Interests

Recursive self-improvement (RSI), coding agents, and long-horizon AI development; verifiable task design and reinforcement learning; inference systems; logical reasoning and tool use.

Education

2024–present **Ph.D. in Computer Science**, Hong Kong University of Science and Technology
HKUST NLP Group; advisor: Prof. Junxian He

2020–2024 **B.Eng. in Automation**, Shanghai Jiao Tong University (IEEE Honor Class)

Research Experience

Apple MLR Research Scientist Intern, Cupertino, USA Apr 2026–Sep 2026
Mentors: Yizhe Zhang and Ruixiang Zhang.

- Developed **RSI-Inference** (working paper, first author), a benchmark for evaluating whether coding agents can autonomously develop and maintain **vLLM and SGLang** in GPU-backed sandboxes, covering two task families:
  - **Model support:** Built tasks requiring native integration of previously unsupported model architectures (e.g., DeepSeek-V4), evaluated through correctness and serving-feature checks.
  - **Optimization:** Built tasks requiring source-code changes to accelerate models already supported by the engine, measuring throughput and latency under fixed deployment settings on H100 and B200 GPUs.
- Sonnet-5.5 and Opus-5.5 led the overall ranking, with reported optimization speedups of **1.68×** and **1.48×**, but model-support pass@1 of only **27.8%** and **16.7%**, respectively. Open-weight baselines lagged behind: DeepSeek-V4.1-Flash achieved **0.0%** support pass@1 and **1.24×** speedup.

MiniMax Research Intern, Shanghai, China Feb 2025–Feb 2026

- **Logical reasoning:** Developed **SynLogic** (first author, NeurIPS 2025), synthesizing rule-verifiable data across 35 logical task types for RL with verifiable rewards. SynLogic-32B outperformed DeepSeek-R1-Distill-Qwen-32B on BBEH (**25.5% vs. 19.2%**). This data and RL work contributed to the **MiniMax-M1** release.
- **Search agents:** Developed **WebExplorer** (first author), synthesizing challenging search tasks through web exploration and query evolution, and explored RL training. WebExplorer-8B surpassed WebSailor-72B on BrowseComp-en/zh (Avg@4: **15.7% vs. 12.0%** / **32.0% vs. 30.1%**). The data was used in the **MiniMax-M2** release and formed the basis for subsequent **MiniMax-M3** search-data improvements.

Tencent WXG Research Intern, Shanghai, China Jun 2024–Sep 2024
Mentor: Zifei Shan.

- Investigated vision-encoder bottlenecks in VLM chart understanding and trained improved encoders with contrastive learning. First-author paper: **On the Perception Bottleneck of VLMs for Chart Understanding** (Findings of EMNLP 2025).

Shanghai AI Lab Research Intern, Shanghai, China Jun 2023–Dec 2023
Mentor: Prof. Yu Cheng.

- Investigated a universal truthfulness hyperplane using probes trained on 40+ datasets; found that data diversity improves cross-dataset generalization. First-author paper: **On the Universal Truthfulness Hyperplane Inside LLMs** (EMNLP 2024).

Publications

* denotes equal contribution.

[1] MiniMax Team (including **Junteng Liu**). The MiniMax-M2 Series: Mini Activations Unleashing Max Real-World Intelligence. arXiv:2605.26494, 2026. [Paper]

[2] Junlong Li, Wenshuo Zhao, Jian Zhao, et al. (including **Junteng Liu**). The Tool Decathlon: Benchmarking Language Agents for Diverse, Realistic, and Long-Horizon Task Execution. ICLR 2026. [Paper]

[3] Wei Liu, Ruochen Zhou, Yiyun Deng, Yuzhen Huang, **Junteng Liu**, et al. Learn to Reason Efficiently with Adaptive Length-Based Reward Shaping. ICLR 2026. [Paper]

[4] Aili Chen, Chi Zhang, **Junteng Liu**, et al. DIVE: Scaling Diversity in Agentic Task Synthesis for Generalizable Tool Use. ICML 2026. [Paper]

[5] **Junteng Liu***, Yunji Li*, Chi Zhang, et al. WebExplorer: Explore and Evolve for Training Long-Horizon Web Agents. arXiv:2509.06501, 2025. [Paper]

[6] MiniMax Team (including **Junteng Liu**). MiniMax-M1: Scaling Test-Time Compute Efficiently with Lightning Attention. arXiv:2506.13585, 2025. [Paper]

[7] **Junteng Liu**, Yuanxiang Fan, Zhuo Jiang, et al. SynLogic: Synthesizing Verifiable Reasoning Data at Scale for Learning Logical Reasoning and Beyond. NeurIPS 2025. [Paper]

[8] **Junteng Liu**, Weihao Zeng, Xiwen Zhang, et al. On the Perception Bottleneck of VLMs for Chart Understanding. Findings of EMNLP 2025. [Paper]

[9] **Junteng Liu**, Shiqi Chen, Yu Cheng, Junxian He. On the Universal Truthfulness Hyperplane Inside LLMs. EMNLP 2024. [Paper]

[10] Shiqi Chen, Miao Xiong, **Junteng Liu**, et al. In-Context Sharpness as Alerts: An Inner Representation Perspective for Hallucination Mitigation. ICML 2024. [Paper]

[11] Jinghan Zhang, Shiqi Chen, **Junteng Liu**, Junxian He. Composing Parameter-Efficient Modules with Arithmetic Operation. NeurIPS 2023. [Paper]

[12] Yuzhen Huang, Yuzhuo Bai, Zhihao Zhu, et al. (including **Junteng Liu**). C-Eval: A Multi-Level Multi-Discipline Chinese Evaluation Suite for Foundation Models. NeurIPS 2023. [Paper]

Honors & Awards

Zhiyuan Honor Scholarship, Shanghai Jiao Tong University (2020–2024).